Machine-Learning Design of Broadband Rayleigh-Wave Carpet Cloaks under Microstructure Realisability Constraints

David Aznaurov*[a], Davit Piliposyan[a], Sébastien Guenneau[b]

[a]Mathematical Modelling and Machine Learning Lab, Institute of Mechanics, Armenia
[b]Department of Physics, Imperial College London

*Presenting author: david@mechins.sci.am

Problem

Challenge

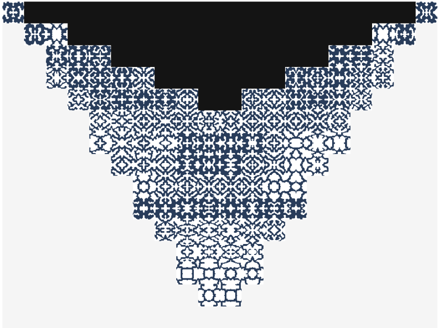
Cloak an arbitrary surface obstacle from Rayleigh waves.

Current Gap

No general solution has yet demonstrated ideal realizable carpet cloaking.

Result

A cement-based proof of concept for single-frequency and broadband cloaking.

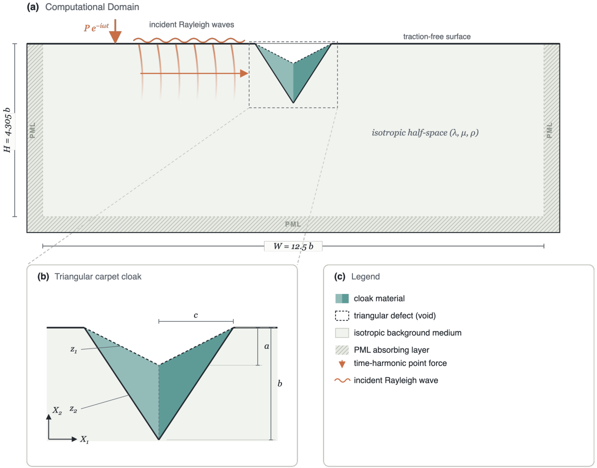


A cloak made of *one* material, patterned cell by cell.

Hiding from a surface wave

The goal

- A **Rayleigh wave** rolls along the free surface of a half-space. A surface *inclusion* scatters it and sensibly affects the displacement field beyond the defect.
- A **carpet cloak** is a coating that makes the surface wave behave as if the inclusion were not there.



Transformation elasticity: the exact cloak

In the homogeneous reference (virtual) half-space $X = (X_1, X_2)$, the in-plane displacement $U = (U_1, U_2)$ obeys the Navier elastodynamic equation

$$\nabla_X \cdot (C : \nabla_X U) = \rho\, U_{tt},$$

for a time-harmonic wave $U(X,t) = U(X)\, e^{i\omega t}$. A coordinate map $x = \Xi(X)$, $F = \nabla_X x$, $J = \det F$, and the gauge $U(\Xi(X)) = u(x)$ carry it to $\nabla_x \cdot (c^{eff} : \nabla_x u) = \rho^{eff} u_{tt}$ with the **Brun–Guenneau–Movchan** push-forward:

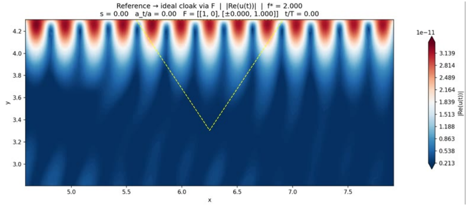
$$c^{eff}_{ijkl} = J^{-1}\, C_{IjKl}\, F_{iI}\, F_{kK}, \qquad \rho^{eff} = \rho\, J^{-1}.$$

The obstruction

The transformed tensor keeps the *major* symmetry but breaks the *minor* ones,

$$c^{eff}_{ijkl} = c^{eff}_{klij}, \qquad c^{eff}_{ijkl} \neq c^{eff}_{jikl}.$$

Hence the ideal cloak is a **polar (Cosserat)** medium with a non-symmetric stress.



The map Ξ morphing the flat reference half-space into the notched one.

Objective

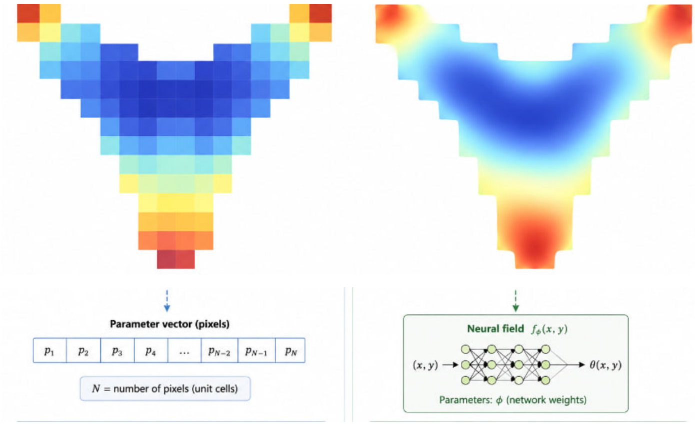
Cast carpet-cloak design as a **PDE-constrained optimisation** solved by a coordinate **neural field** through a **differentiable finite-element** wave solver, and carry the design all the way to a fabricable single-material microstructure.

1. **Neural-field material optimisation** – the main pipeline (Sec. 3).
2. **Unit-cell inverse design** with a neural field.
3. **Guided-diffusion** generation of unit cells.
4. **Latent-space search** over a learned design manifold.

Unconstrained design $\longrightarrow \eta = 0.99$ (near-ideal)

Best *fabricable* design $\longrightarrow \approx 97\%$ of ideal

First microstructured Rayleigh-wave cloak from a common material.



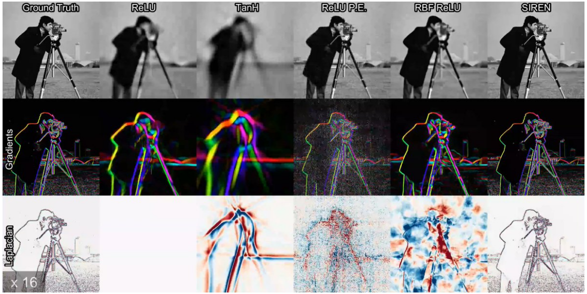
A pixel grid stores one value per unit cell; the neural field $f_\phi(x, y)$ stores the whole material as network weights — resolution-free and smooth by construction.

- **Resolution-free**: query anywhere, e.g. at FEM quadrature points.
- **Compact**: $\mathcal{O}(10^3)$ weights for a whole material field.
- **Differentiable**: $\partial(\text{field})/\partial\theta$ in closed form.
- **Smooth by construction**: the architecture, not a penalty, sets the bandwidth.

A small MLP that maps a **coordinate** to a **field value**,

$$\Phi_\theta : (x, y) \mapsto \text{field},$$

trained until it *is* the field. The weights θ, not a pixel grid, are the degrees of freedom.



Same target, same budget, different activation/encoding (Sitzmann et al., 2020).

Elastic cloaking optimization formulation

Design class: orthotropic Cauchy (4CM)

Square D_2-symmetric cells force $C_{16} = C_{26} = 0$:

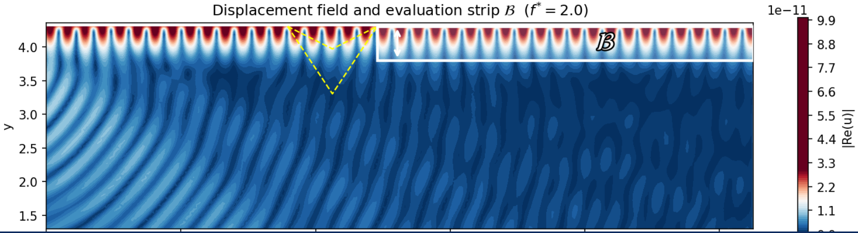
$$C_{4CM} = \begin{bmatrix} C_{11} & C_{12} & 0 \\ C_{12} & C_{22} & 0 \\ 0 & 0 & C_{66} \end{bmatrix} \succ 0, \quad (+\rho).$$

Four moduli + density *per cell* are the design variables.

Objective: phase-invariant magnitude error

$$\mathcal{L}_{cloak}(\theta) = \frac{1}{|\mathcal{B}|} \int_{\mathcal{B}} \left(\frac{|u_{cloak}|}{|u_{ref}|} - 1 \right)^2 dA$$

over a near-surface strip $\mathcal{B}$ downstream of the cloak.



Methodology: one differentiable pipeline

❶ Represent

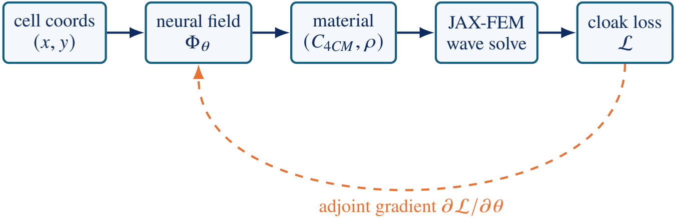
- A coordinate MLP Φ_θ *is* the material field.
- Outputs (C_{4CM}, ρ) per cell.
- Smoothness set by Fourier features.

❷ Solve & differentiate

- Frequency-domain JAX-FEM wave solve.
- Gradients via the implicit **adjoint** of the linear solve.
- End-to-end $\partial\mathcal{L}/\partial\theta$.

❸ Realise

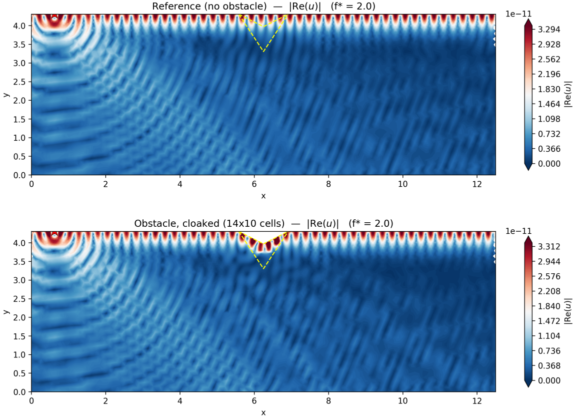
- Project onto a **realisability manifold**.
- Fill each cell with a solid/void microstructure.
- Inverse design / diffusion / latent search.



Results I – one material already beats symmetrisation

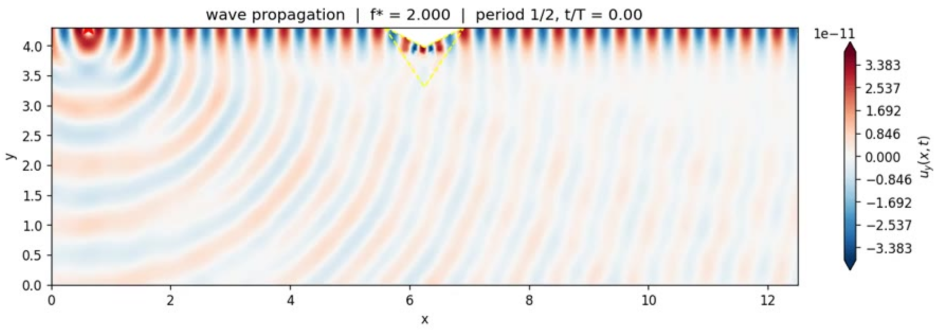
Configuration	Class	η
Obstacle (no cloak)	—	0.37
Symmetrised ideal (2×1) - Chatz. 2023	6CM	0.634
Single optimised (1×1)	4CM	0.857
Optimised L/R pair (2×1)	6CM	0.947
Ideal continuous polar c^{eff}	polar	0.999

Metric: the **cloak ratio** $\eta = \dfrac{\langle |u| \rangle}{\langle |u_{ref}| \rangle} \to 1$ (perfect cloak).



Top⟶bottom: reference (no obstacle), optimised 14×10 cloak ($\eta = 0.99$).

Results I – the cloak in motion



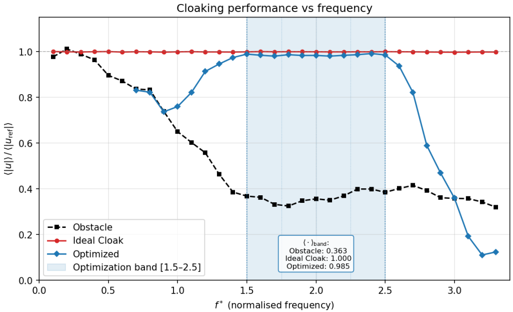
➤ **Takeaway.** The wavefront leaves the optimal cloak intact – no shadow, no reflected front.

Results II – broadband: one design, many frequencies

Train a single neural field against a band of frequencies at once:

$$\mathcal{L}_{band}(\theta) = \frac{1}{\sum_f w_f} \sum_{f \in \mathcal{S}} w_f \, \mathcal{L}_{cloak}(\theta; f).$$

- Band $f^\star \in [1, 3]$, 7 frequencies, equal weights.
- Optional min–max variant flattens the worst case.
- Cost scales linearly in $|\mathcal{S}|$; frequencies solved in parallel.
- Floquet–Bloch check: propagating modes, *no* resonant band-gap trickery.



$\eta(f)$: broadband design (blue) holds ≈ 0.98 across the shaded training band; degrades gracefully outside it.

➤ **Takeaway.** The single-frequency narrowness was a limit of the *objective*, not the material class.

Admissible *by construction*: the decoding map

Raw outputs $r = (r_{11}, r_{22}, r_{66}, r_{12})$ are decoded so every design is a valid Cauchy tensor. Three constraints are *engraved into decoding* (hard, always satisfied); realisability enters as a soft, dataset-based prior.

Positivity

$$C_{ii} = C_{ii}^{(0)} e^{\epsilon r_{ii}}$$

Log-space $\Rightarrow C_{ii} > 0$ for any real r_{ii}.

Bounded anisotropy

$$g_b = \tfrac{1}{2}\log R \, \tanh\!\left(\frac{g}{\tfrac{1}{2}\log R}\right)$$

confines $C_{11}/C_{22} \in [1/R, R]$, $R = 30$.

PD coupling

$$C_{12} = \rho_c \sqrt{C_{11} C_{22}}$$

$\rho_c = \kappa \tanh(r_{12} + \phi_0)$, $\kappa = 0.99$
$\Rightarrow \det > 0$.

Realisability prior (soft, dataset-based)

A homogenised (C_{4CM}, ρ) need not correspond to a manufacturable object. Fit a GMM to $N \sim 10^6$ homogenised binary 50×50 cells and add a log-density penalty

$$\mathcal{L}_{GMM} = \sum_c \max\!\left(0,\ \tau - \log p_{GMM}(C^c_{4CM}, \rho_c)\right),$$

pulling every cell toward the realisable manifold $\mathcal{M}$ without pinning it to one catalogue entry.

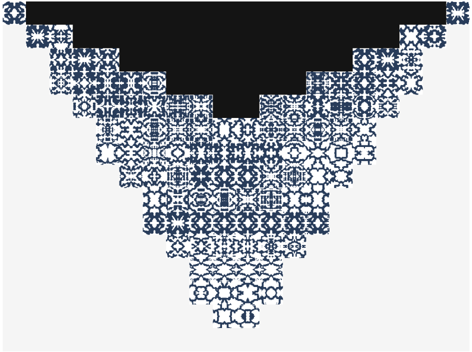
Method 1 – Unit-cell inverse design (neural field)

Given a per-cell target $y = (C_{4CM}, \rho)$, parameterise a *continuous density field* over the 50×50 cell, $\hat{\rho}(x, y) = g_\phi(x, y) \in [0, 1]$, and minimise

$$\min_{\phi} \left\| \mathcal{H}(\hat{\rho}_\phi) - y \right\|^2 + \lambda \mathcal{R}_{conn},$$

where $\mathcal{H}$ is the differentiable periodic-FEM homogenisation operator.

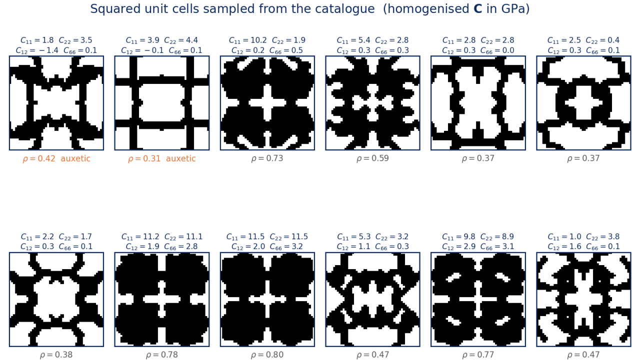
- Soft-to-hard schedule drives $\hat{\rho} \to \{0, 1\}$.
- Connectivity penalty $\mathcal{R}_{conn}$ kills floating islands/voids.
- **Deterministic**: one locally optimal cell per target.



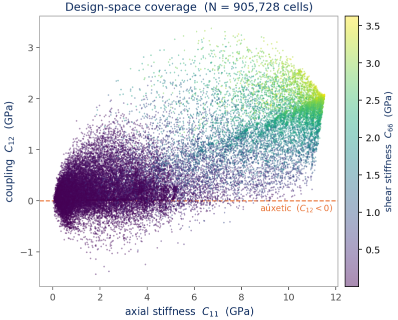
Inverse-design realisation: one cement phase, cell-varying void pattern tiled across the carpet.

Method 2 – Guided-diffusion generation

Design space: dataset of squared cells



Real 50×50 mirror-symmetric (D_2) cells with their homogenised in-plane stiffness [7].



$\sim 10^6$ cells cover a broad (C_{11}, C_{12}, C_{66}) range – including auxetic ($C_{12} < 0$).

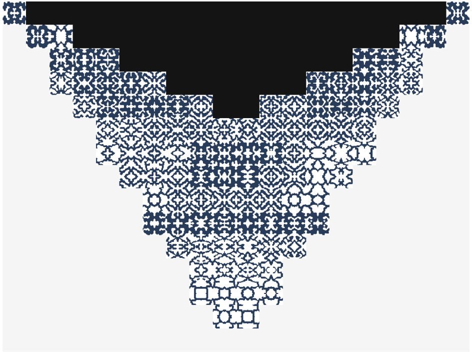
Train a **conditional denoising diffusion** model on the catalogue,

$$\min_{\theta} \mathbb{E}_{t,x_0,\boldsymbol{\epsilon}} \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(x_t, t, y) \right\|^2, \quad y = (C_{4CM}, \rho),$$

then sample the reverse process $p_\theta(x_{t-1} \mid x_t, y)$. Because sampling is stochastic, draw N candidates and keep the closest:

$$x^\star = \arg \min_{n=1..N} \left\| \mathcal{H}(x^{(n)}) - y \right\| \quad (\textbf{best-of-}N).$$

- Prior keeps every sample *on* the manufacturable manifold.
- Explores a diverse set of realisations per target.

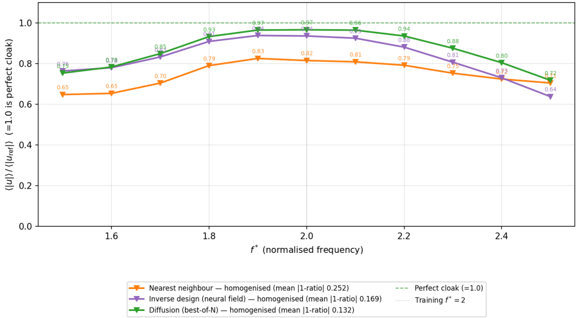


Best-of-N diffusion realisation of the same target field – visibly different, equally manufacturable patterns.

Results III – how much survives fabrication?

Realising the 20×15 cloak (100 cells), $f^\star = 2.0$

Realisation of the 100 cells	η
Continuous 4CM optimum	0.99
Nearest-neighbour snap	0.82
Inverse design (neural field)	0.94
Diffusion (best-of-N)	0.97
Ideal continuous polar c^{eff}	0.999



$\eta(f)$ across the band: diffusion (green) stays closest to the perfect-cloak line; smallest mean distortion.

➤ **Takeaway.** Better per-cell matching ⟹ better cloak: best-of-N diffusion recovers **97%** of the ideal, from fabricable single-material cells.

Conclusions

- A single **differentiable pipeline** – coordinate neural field + JAX-FEM adjoint – designs Rayleigh-wave carpet cloaks inside the realisable Cauchy class.
- **Near-perfect** single-frequency ($\eta = 0.99$) and **broadband** (≈ 0.98 over a band) cloaking.
- Four ML routes to a real structure – neural-field optimisation, inverse design, guided diffusion, latent search – with **diffusion recovering** $\approx 97\%$ of ideal performance from a *single* constituent.
- Demonstrated that Rayleigh-wave carpet cloaks do **not** require assymetric materials.

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Thank You

Acknowledgements

The authors gratefully acknowledge Danila Rukhovich (Google) for support. Code, dataset, and the differentiable JAX-FEM pipeline available on GitHub https://github.com/M3LLab/nn-cloaking